A cavity-enhanced large-period optical lattice with graded-phase modes for 10^5 to 10^6 cesium atoms

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Abstract

Static meta-optic tweezer arrays for 10^6 cesium atoms need about 1.4 kW of trapping light at 1075 nm, which makes laser power, heat in the metasurface and damage by absorbing defects the limits on scaling. We show that a build-up cavity cannot reduce the power of such an array tenfold, because every intracavity focusing element is passed at least twice per round trip; the enhancement is bounded at 3.2. We then design a cavity-enhanced lattice with a 4 µm pitch, in which each axis is a self-intersecting ring whose flat-top sheet mode is set by two graded-phase mirrors and made non-degenerate by a 0.2× intracavity telescope. Fox–Li simulations of the ring give a barrier depth uniform to 0.23 % rms over 10^6 sites, and the three rings need 3.5 W of input at an enhancement of 300 (0.69 W at 1500), against 1.04 kW for the same lattice in free space. Mirror figure errors of $\lambda/1000$ rms on every mirror give 2.0 to 4.8 % before correction; feedback on depths measured with the atoms restores 0.22 to 0.54 % in four of six runs. The constraints move to mirror figure, alignment stability, surface cleanliness at 4.5 to 19 kW/cm^2 (up to 170 kW/cm^2 on the vertical ring as folded), and mirror vibration below 0.32 pm/$\sqrt{\text{Hz}}$ at the 20 kHz trap frequency.

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1 Introduction

A neutral-atom array of 10^5 to 10^6 sites, trapped by meta-optic tweezers, needs trapping light in proportion to the number of sites. With the meta-hologram of this project (143 mK site per watt) a 0.2 mK trap at every one of 10^6 sites needs 1.40 kW at the optics input, and the lenslet array 2.50 kW. At this power three limits act together: (i) the laser power itself; (ii) the heat deposited in the metasurface and its substrate; and (iii) damage, and thermal lensing, caused by absorbing defects on surfaces that carry the full beam. A build-up cavity is the standard way to reduce the input power, because the light is reused many times. The question we address is whether a combination of a cavity and meta-optics can reduce the power by an order of magnitude at 10^6 sites, and which design and simulations support it.

Two experiments are the closest precedents. Park et al. built a two-cavity Sr lattice with finesse 5025 and a power enhancement of 1132 at 914 nm, with 489 µm waists and about 92 W circulating in a monolithic ULE assembly in vacuum [1]. Norcia et al. imaged 1225 Yb sites in two self-intersecting cavities (finesse 2900 and 3000, waists 268 and 183 µm) with roughly two orders of magnitude less power than tweezers and at most 50 ppm loss per mirror [2]. Both used Gaussian modes over arrays of 0.1 to 1 mm. Neither shaped a flat mode over several millimetres, and neither addressed the near-degeneracy of the transverse modes that such a mode brings.

In this work, we (i) bound the enhancement of a tweezer array placed inside a cavity (Section 2); (ii) design a three-axis lattice of self-intersecting rings with graded-phase mirrors and an intracavity telescope, and fit it around the atoms with room for the imaging objective, a

MOT and further beams (Sections 3 and 8); (iii) give its power and loss budget (Section 4); (iv) simulate its flat-top mode, tolerances and correction with a Fox–Li model (Sections 5 and 6); (v) evaluate the intensities, defect limits and thermal deformation on the mirrors (Section 7); and (vi) derive the noise requirements set by heating (Section 9). Section 10 separates what is shown from what depends on the model, and lists the next steps. A parameter summary, the commands that reproduce every result, and the provenance of every number are given in the appendices.

2 Why the tweezer array cannot be cavity-enhanced tenfold

A ring cavity with input transmission T and all other round-trip losses L has the on-resonance power enhancement

$$G = \frac{T}{(1-r)^2}, \qquad r = \sqrt{(1-T)(1-L)}, \tag{1}$$

with the maximum $G_{\max} = 1/L$ at $T = L$. Here, G is the ratio of circulating to input power. For $G = 10$ the round-trip loss must stay below 10 %. An NA 0.5 tweezer array is not an eigenmode of a pair of mirrors, so the focusing optic must sit inside the cavity, and it is passed at least twice per round trip. Table 1 gives the resulting bounds.

Table 1: Largest enhancement of a cavity that contains the focusing optic of a tweezer array. Efficiencies of the meta-optics from this project's simulations; the refractive and relay values are assumptions.

Intracavity focusing element	Passes per round trip	$G_{\max}$
Meta-hologram, efficiency 0.83	2	3.2
Pair of meta-objectives, 0.80 each	4	1.7
Pair of refractive objectives, 0.95 each [VALUE?]	4	5.4
Ideal AR relay, 0.995 per pass [VALUE?]	4	50

Only the last line reaches 10, and it is not a meta-optic. A degenerate concentric cavity avoids intracavity lenses: two spherical mirrors share their centre at the atoms and image every focus onto itself. However, a ray trace of one reflection (NA 0.5, mirror radius 30 or 50 mm, piston, tilt and focus removed) gives a round-trip wavefront error of 0.031 or 0.019 waves rms at a 100 µm field, 0.28 or 0.17 at 300 µm, and 3.1 or 1.9 at 1 mm. Each site must stay within a fraction of the linewidth $2\pi/F$, so the finesse F is limited to 3 to 5 at 100 µm and below 1 at 300 µm. The mirrors would also fill both NA 0.5 cones that imaging needs. Retro-reflecting the tweezer light through a second objective onto a mirror gives at most $(1 + \sqrt{\eta_{\text{return}}})^2$, i.e. 3.2 for the meta-objective ($\eta_{\text{return}} = 0.8^2 \times 0.99$) and 3.8 for a refractive one, and it forms a $\lambda/2$ axial standing wave in every trap.

Thus the order-of-magnitude saving must come from a different trap geometry: one in which each photon serves many sites, because each lattice beam crosses every row of the array, and whose mode can be supported by mirrors alone.

3 Architecture

3.1 Three self-intersecting rings

The lattice is formed by three build-up cavities, one per axis, mutually detuned by at least 100 MHz so that their interference averages out (0.36 ppm of the laser frequency; the resulting site shift is 0.7 nm at the array edge). Each in-plane axis is a self-intersecting ring. Its mode crosses the atom plane twice, on arms A and B, at a full angle 2θ with

$$\sin\theta = \frac{\lambda}{2a}, \tag{2}$$

so that $\theta = 7.72°$ for a period $a = 4\,\mu\text{m}$ at $\lambda = 1075\,\text{nm}$. Both crossing beams travel in the same direction, so the only fringe period is a; a standing-wave cavity would add $\lambda/2$ fringes along the beams. The two beams of one axis are the same circulating mode, so their relative phase needs no separate lock. The polarisation is vertical, perpendicular to the crossing plane, which gives full contrast.[1] The y-lattice ring has its arms along x, and the x-lattice ring is the same ring rotated by 90°.

The mode is a sheet. In the plane it is a super-Gaussian flat-top with amplitude $\exp(-|u/w|^{20})$, whose intensity is at least 0.99 of the flat level over the array's transverse range $|u| \le (S/2)(\cos\theta + \sin\theta) = 2.25\,\text{mm}$. Here, u is the transverse coordinate of the beam and $S = 4.0\,\text{mm}$ the side of a 1000×1000 array. This gives $w = 2.93\,\text{mm}$ and an equivalent width $W_{\text{eq}} = 5.51\,\text{mm}$. Vertically the sheet is Gaussian with a 75 µm half-thickness w_z at the crossing, whose Rayleigh range of 16.4 mm keeps the depth roll-off along the beams within 1 % over the array.

3.2 Graded-phase mirrors and the telescope

Each half trip, from the crossing on arm A back to the crossing on arm B, is symmetric about a graded-phase mirror G. With the target flat-top F at the crossing, the mirror phase

$$\phi_G = -2\arg(KF) \tag{3}$$

returns the conjugate field $(KF)^*$, and the symmetric return path then gives $F^* = F$. Here, K is the propagation operator from the crossing to G. Hence F is an exact eigenmode of any lossless symmetric half trip; this is the graded-phase resonator of Bélanger and Paré [3]. Section 5 shows that a plain ring of this kind is nearly degenerate. The adopted design therefore puts a 0.2× Keplerian telescope of cylindrical mirrors in each half trip: a primary P1 ($f = 100\,\text{mm}$), a secondary P2 ($f = 20\,\text{mm}$) 120 mm further, and G 100 mm beyond P2. In the layout of Fig. 1 the primary is 70 mm from the crossing, because at 50 mm the mirrors of the two in-plane rings overlap by 4.2 mm. The ring then has a half trip of 290 mm, a round trip of 1.16 m, a free spectral range of 258 MHz, 10 reflections per round trip, and angles of incidence of 15° (P1), 44° (P2) and 24° (G). In the Fox–Li model it is equivalent to the 50 mm ring used for most of Sections 5 and 6: depth uniform to 0.23 % rms, fundamental loss 1.6×10^{-8} per round trip, nearest competing mode at 3.3×10^{-3} loss (against 1.1×10^{-2} at 50 mm), and $\lambda/1000$ errors on all ten mirrors give 2.0 and 4.0 % uncorrected and 0.42 and 0.75 % after six rounds of atom feedback.

The vertical Gaussian is set by a vertical focal length on P1 and a vertical radius on G. For the 70 mm ring, 40 mm and 143 mm give the 75 µm waist with a half-trip stability parameter $m = -0.24$, a round-trip Gouy phase of $-0.99\,\text{rad}$, beam half-heights of 0.33 mm on P1 and 0.55 mm on G, and a 0.32 % waist change per 0.1 % radius error.

3.3 Vertical confinement and the vertical ring

The in-plane sheets alone give only 0.7 kHz vertically (Gaussian profile of 75 µm, 0.4 mK at the site). A third ring therefore crosses at $2\theta_z$ in a vertical plane (period 8 µm, $\theta_z = 3.85°$) and propagates along the array diagonal. Its beams walk off vertically by ±0.19 mm across the array, so they must be flat-topped vertically as well: 6.9 mm wide, with an equivalent height of 0.57 mm and an order-10 edge, because an order-20 edge has a 4×10^{-4} design loss at this small size. This axis takes 785 W of the 989 W circulating at 10^6 sites and 0.2 mK.

The vertical ring (Fig. 3) is planar, in the vertical plane through the atoms that contains the array diagonal. A planar ring has no image or polarisation rotation, so the flat-top axis and the polarisation stay horizontal, perpendicular to the ring plane, and the cylindrical telescope mirrors curve perpendicular to their plane of incidence, where a parabolic cylinder is an exact

[1] Horizontal polarisation would allow Brewster-angle windows, at a contrast $\cos 2\theta = 0.964$.

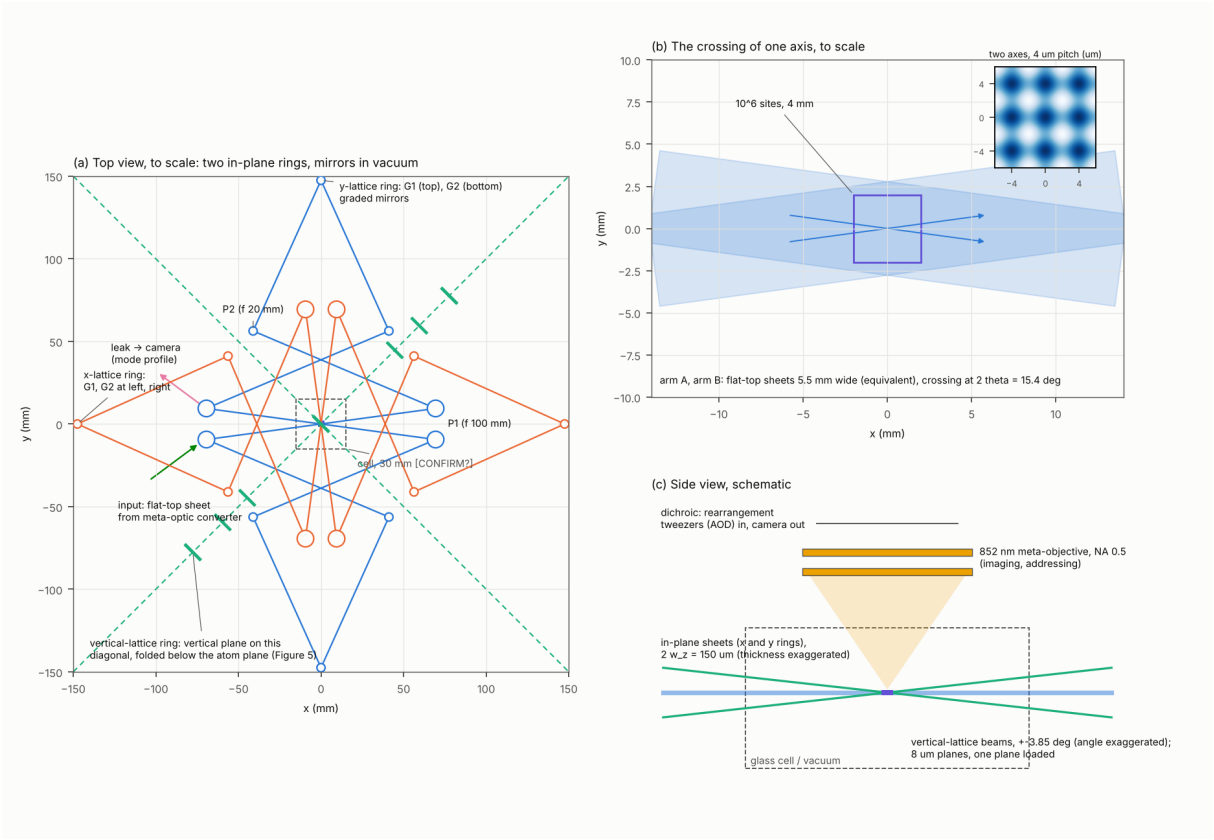


Figure 1: Arrangement of the in-plane rings. (a) Top view to scale: the y-lattice ring (blue) and the x-lattice ring (orange), each a self-intersecting ring with a 0.2× cylindrical telescope (primary P1, $f = 100\,\mathrm{mm}$, 70 mm from the crossing; secondary P2, $f = 20\,\mathrm{mm}$) and two graded mirrors G1, G2; circles are the mirror apertures (10 and 5 mm). Dashed diagonals: the corridor for the vertical-lattice ring, whose fold geometry is shown in Fig. 3 (this sketch predates it). Green: the flat-top input sheet from an extracavity meta-optic converter; pink: the light leaked by a fold mirror to the mode camera. (b) The crossing of one axis to scale: the two flat-top sheets (equivalent width 5.5 mm, $2\theta = 15.4°$) cover the 4 mm array; inset, the computed intensity of the two incoherent in-plane axes (plane waves, 4 µm pitch). (c) Side view, schematic, not to scale: the in-plane sheets, the two vertical-lattice beams (±3.85°, 8 µm planes) and the 852 nm NA 0.5 meta-objective above the cell.

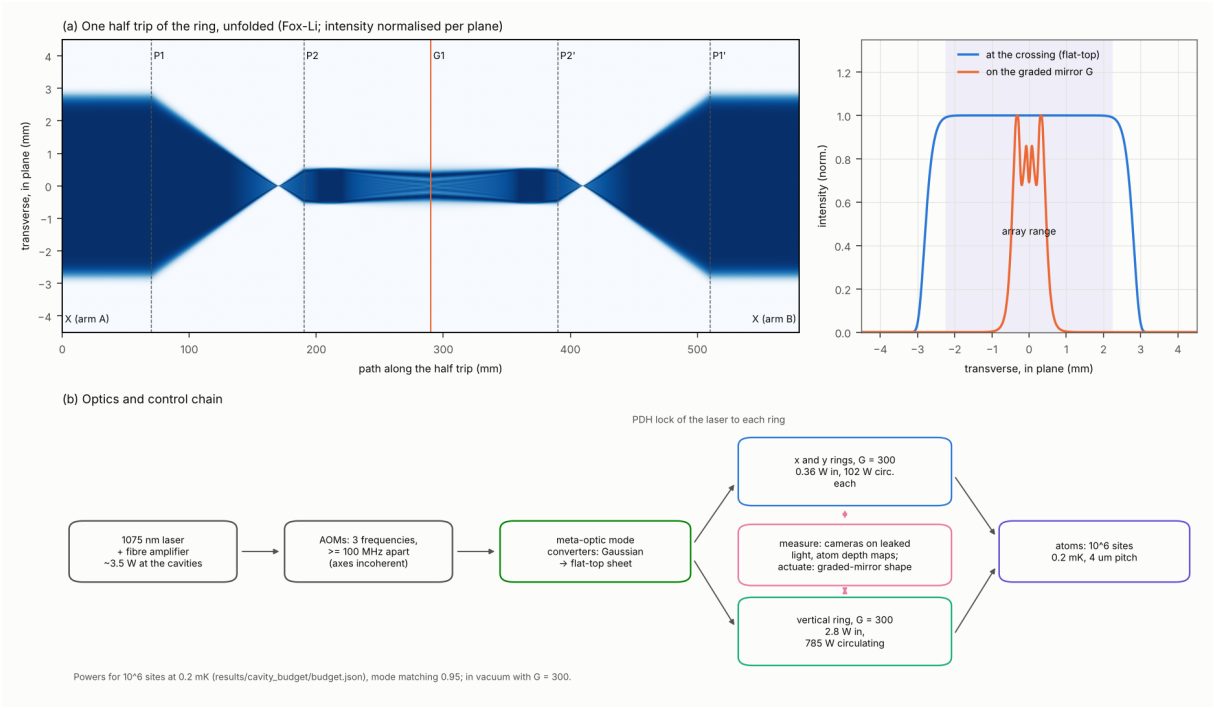


Figure 2: (a) The beam along one unfolded half trip of the 70 mm ring, from the crossing on arm A through P1, P2, G1, P2′, P1′ to the crossing on arm B (Fox–Li, 8192 points; intensity normalised in each plane). The flat-top is compressed fivefold by the telescope, passes a line focus between P1 and P2, and is shaped back by G1. Right: the profiles at the crossing and on G. (b) Optics and control chain for 10^6 sites at 0.2 mK with $G = 300$ and mode matching 0.95: 0.36 W into each in-plane ring (102 W circulating) and 2.8 W into the vertical ring (785 W circulating). The graded-mirror correction of Section 6 closes the loop from the mode cameras and the atom depth maps.

cylindrical lens of focal length $R/(2\cos\mathrm{AOI})$. Both half trips fold below the atom plane: nothing of the ring rises above 9.4 mm, which leaves the space above the atoms to the objective. Because the crossing half-angle is small, the exit mirrors of the two half trips sit close together (P1 and P3, 14.8 mm apart vertically at 110 mm), and with an angle of incidence of at most 45° every fold must turn the beam by at least 90°.

Our choice of telescope for this ring is a compromise between two considerations. First, with the in-plane telescope ($f = 100$ mm, 0.2×) the best layouts found needed a 60 mm graded-mirror distance, which leaves a competing mode 0.07 rad away with 10^{-5} loss and gives 7.7 to 69 % non-uniformity for $\lambda/1000$ errors (exploratory run). Second, a longer telescope must still clear the in-plane rings in three dimensions. A search over the primary focal length, demagnification, distances and fold directions selects a primary of $f = 160$ mm, a secondary of $f = 24$ mm (0.15×), the primary 110 mm from the atoms and G 120 mm behind the secondary: half trip 414 mm, round trip 1.66 m, free spectral range 181 MHz. Half trip 1 folds back under the atoms (its folded beam passes 30.6 mm from them) to G1, 158 mm below the atoms; half trip 2 folds down to 190 mm depth and back up to G2, 104 mm below the atoms. The clearances are 11.4 mm between mirror faces, 11.1 mm between beams and foreign mirrors, and 37.8 mm and 35.8 mm to the in-plane rings' mirrors and beams; the angles of incidence are 8° to 45°. The effective propagation behind the telescope, $L/M^2 = 5.3$ m, is twice that of the in-plane rings (2.5 m), which partly compensates the wider beam.

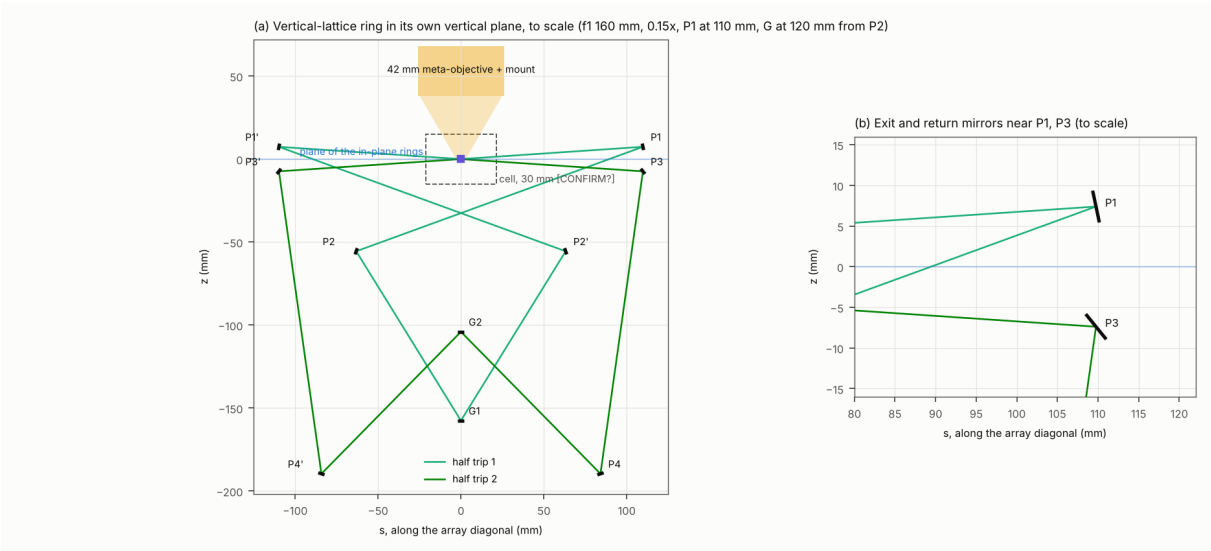


Figure 3: The vertical-lattice ring in its vertical plane through the array diagonal, to scale (s along the diagonal, z up). Half trip 1 (light green) leaves the crossing on the upper arm, folds at P1 back under the atoms to P2, the graded mirror G1 on the vertical axis 158 mm below the atoms, P2′ and P1′, and returns on the lower arm; half trip 2 (dark green) folds via P3, P4, G2 (104 mm below), P4′, P3′. Black bars: mirror faces (4 mm strips in the ring plane; widths along the horizontal flat-top axis 12 and 5 mm). Telescope $f_1 = 160$ mm, $f_2 = 24$ mm. Yellow: the 42 mm imaging objective with its mount [CONFIRM?] and the NA 0.5 light cone. (b) The exit and return mirrors near P1, P3.

3.4 Role of the meta-optics

None of the meta-optics sees the circulating power. They provide (i) the input mode converters, from a Gaussian to the flat-top sheet; a Gaussian input couples 0.918 of its power into the flat-top mode, and a flat-top converter is assumed to reach 0.95 [VALUE?]; (ii) the 852 nm imaging and addressing objective; and (iii) the tweezers for loading and rearrangement.

4 Power and loss budget

Each crossing beam needs the peak intensity

$$I_1 = \frac{U}{4c\kappa}, \tag{4}$$

where U is the barrier depth and $\kappa = 2.53 \times 10^{-13}\,\mathrm{K}$ per $\mathrm{W/m^2}$ for Cs at 1075 nm (polarisability 1125 a.u.). For $U = 0.2\,\mathrm{mK}$ this is $19.8\,\mathrm{kW/cm^2}$. The circulating power per in-plane axis is $I_1 W_{\mathrm{eq}} \sqrt{\pi/2}\, w_z$, independent of the number of beams per axis because one mode supplies both. Thus the in-plane power scales as the array side, $\sqrt{N}$, against N for tweezers. The vertical axis scales close to N, because its beams must also cover the vertical walk-off (Table 2).

Table 2: Circulating power of the three lattice axes and the input power of hologram tweezers of the same depth. The hologram column reproduces the 210 W of the earlier 10^5-site study at 0.3 mK.

N, barrier	Per in-plane axis	Vertical axis	Total circulating	Hologram tweezers
10^5, 0.2 mK	32 W	78 W	143 W	140 W
10^5, 0.3 mK	48 W	117 W	214 W	210 W
10^6, 0.2 mK	102 W	785 W	989 W	1399 W
10^6, 0.3 mK	154 W	1177 W	1484 W	2098 W

The round-trip loss, with 10 reflections of 50 ppm (the per-mirror bound of Ref. [2]), and the resulting enhancement are given in Table 3.

Table 3: Round-trip loss and enhancement for the placements considered. The design point is the over-coupled input coupler $T = 3L$.

Placement	Round-trip loss L	$G_{\max} = 1/L$	G at $T = 3L$ (finesse)
Mirrors in vacuum	5.0×10^{-4}	2000	1500 (3140)
In vacuum, graded mirrors $+10^{-3}$ each	2.5×10^{-3}	400	300 (627)
In vacuum, graded mirrors $+10^{-2}$ each (metasurface)	2.0×10^{-2}	49	36 (75)
Through the cell, 8 AR surfaces at 2.5×10^{-4} [VALUE?]	2.5×10^{-3}	398	298 (623)
Through the cell, 8 AR surfaces at 10^{-3} [VALUE?]	8.5×10^{-3}	118	88 (183)
Through the cell, 8 Brewster surfaces at 10^{-4} [VALUE?]	1.3×10^{-3}	760	570 (1192)

We take $T = 3L$ as the design point. It gives 0.75 of the maximum enhancement at half the finesse, and the enhancement then depends only weakly on the exact loss; Section 5 shows that the depth tolerances do not depend on the finesse between 47 and 7000. A further reason to stay well below $G_{\max}$ is the cavity linewidth: 413 kHz at $G = 300$ and 82 kHz at $G = 1500$ for the 258 MHz free spectral range. The linewidth converts laser frequency noise into intensity noise, which Norcia et al. identify as a heating source (190 ms time constant) [2], and Section 9 quantifies the cost. We therefore recommend $G \approx 300$ with mirrors in vacuum, i.e. 3.5 W for 10^6 sites at 0.2 mK, and use $G = 1500$ (0.69 W) as the upper bound. At equal radial trap frequency instead of equal depth (41 kHz, which needs a 0.86 mK barrier) the in-vacuum lattice needs 1.4 W at $G = 1500$ and 6.9 W through good AR windows.

The vertical axis dominates at 10^6 sites. Three options reduce it, none simulated here: (i) a larger vertical period, since the power scales as $1/a_z$ (as does the vertical trap frequency); (ii) propagation along an array side instead of the diagonal, at about half the power but with less clearance to one in-plane ring; and (iii) a lower vertical barrier, if imaging tolerates it. Without the vertical axis the in-plane lattices need 205 W circulating at 10^6 sites, i.e. 0.7 W of input at $G = 300$.

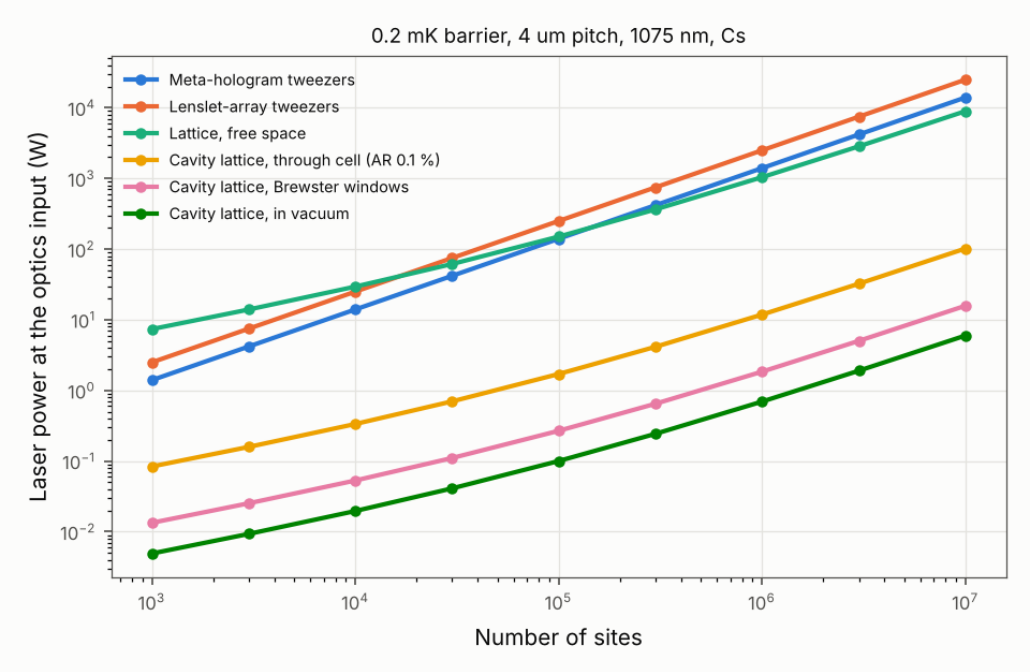


Figure 4: Laser power at the input of the trapping optics versus the number of sites, for a 0.2 mK barrier at 1075 nm and 4 µm pitch. Tweezer lines: 143 (hologram) and 80 (lenslet) mK site per watt from earlier simulations of this project. Lattice lines: two in-plane axes and one vertical axis (8 µm period), flat-top times Gaussian sheets ($w_z = 75$ µm), mode matching 0.95; cavity enhancement for $T = 3L$ with 10 reflections of 50 ppm per round trip, plus 8 AR surfaces of 0.1 % (through the cell) or 8 Brewster surfaces of 10^{-4}.

5 The flat-top cavity mode

5.1 Method

We model the horizontal profile with a one-dimensional strip Fox–Li model on 8192 points over 24 mm (2.9 µm sampling), with exact scalar angular-spectrum propagation, hard apertures at every mirror and the graded-phase design computed through the actual elements. The vertical profile is separable; a separable two-dimensional propagation agrees with the product of one-dimensional propagations to 2×10^{-3}. The driven steady state $(I - re^{i\psi}R)E = tE_{\text{in}}$ is solved by GMRES. Here, R is the round-trip operator, ψ the round-trip phase set by the laser, and E_{in} the input field. The laser is locked to the ψ of maximum circulating power, found by re-centring on the round-trip phase of the circulating field followed by a golden-section search. The fields of the two arms are propagated across the array, combined with the vertical Gaussian (width and Gouy phase) and sampled at 101×101 sites spanning the 10^6-site array. The metrics are the rms and minimum-to-mean ratio of the barrier depth, and the site displacement along the fringe normal with and without its affine part (a uniform pitch change or rotation is a calibration, not an error). Unless stated, the input is the target flat-top, $G = 300$ and the lumped loss is 3×10^{-4}. The model reproduces the strip Gaussian mode of a plane-concave design and its first-order Gouy phase to 10^{-4}, and the driven enhancement against Eq. (1) to 10^{-4}. The results do not change between 4096, 8192 and 16384 points (depth rms 0.23148, 0.23149 and 0.231 49 %).

5.2 Ideal designs and the degeneracy of the plain ring

Every design reproduces the target. Over 10^6 sites the barrier depth is uniform to 0.23 % rms (minimum over mean 0.990, set by the 1 % design tolerance at the flat edge). The site displacement is 6 nm rms, all of it affine (residual 0.03 nm), and it comes from the differential

Gouy phase of the two sheets. Table 4 compares the designs.

Table 4: Ideal designs. Spectra by full diagonalisation of the round trip on 4096 points. The graded mirror column gives its peak-to-valley phase and maximum deflection angle.

Design	Fresnel number $w^2/(\lambda p)$	Fundamental loss per round trip	Nearest competing mode (phase, loss)	Graded mirror PV, deflection
Plane folds, $p = 150$ mm	53	8×10^{-10}	0.017 rad, 2×10^{-9}	5.9 rad, 0.004
Plane folds, $p = 300$ mm	27	1×10^{-9}	0.033 rad, 6×10^{-8}	9.0 rad, 0.004
Plane folds, $p = 600$ mm	13	3×10^{-6}	0.067 rad, 8×10^{-6}	13 rad, 0.003
Telescope 0.2×, Keplerian	–	1.4×10^{-8}	all modes lose $\geq 1.1 \times 10^{-2}$	24 rad, 0.013
Telescope 0.1×, Keplerian	–	2.3×10^{-8}	all modes lose $\geq 1.4 \times 10^{-2}$	36 rad, 0.026
Telescope 0.2×, Galilean	–	2.6×10^{-6}	0.33 rad, 1.5×10^{-3}	30 rad, 0.013

With plane folds the mode spacing is about $\lambda p/w^2$, the inverse Fresnel number of the flat mode: 0.017 rad at $p = 150$ mm and 0.067 rad at 600 mm, with no loss difference. A perturbation of the round-trip phase $\delta\phi$ mixes the nearest mode into the fundamental with an amplitude of about $\delta\phi/0.017$. A $\lambda/1000$ surface error, i.e. 12.6 mrad of reflected phase, is then an order-one admixture. This is the flat-top counterpart of the w^4 alignment sensitivity of large Gaussian modes noted by Park et al. [1]. Curved folds do not help: a single cylindrical fold raised the spacing to at most 0.025 rad, and Fourier-transform configurations made it smaller (exploratory runs). A demagnifying telescope of magnification M makes the remaining path L to the graded mirror act, seen from the crossing, like a path L/M^2. The mode on G then has a Fresnel number near 1, the regime of strong mode discrimination of the graded-phase resonator. In the 0.2× design every competing mode loses at least 1.1 % per round trip, a million times the loss of the fundamental. The edge order is a compromise: orders 10 and 40 give the same uniformity (0.24 and 0.23 %) at equivalent widths of 6.78 and 4.98 mm, but in the telescope ring order 40 raises the design loss from 1.4×10^{-8} to 3.4×10^{-5} and brings a competing mode to 4.7×10^{-3} loss, so we keep order 20.

5.3 Tolerances

Table 5 gives the barrier-depth non-uniformity for figure errors, curvature errors and tilts, as the median over two random realisations and correlation lengths of 0.5 and 2 mm referred to the crossing. Figure errors are rms surface heights over the beam, with reflected phase $4\pi h/\lambda$; both graded mirrors are perturbed independently, and tilts and curvature errors are on one graded mirror. The P1 rows put the same error on the incoming primary of both half trips; Section 6 treats independent errors on all ten mirrors.

Four results follow. (i) The telescope reduces the sensitivity to graded-mirror figure 6 to 11 fold, and to graded-mirror tilt 20 to 60 fold. (ii) The large telescope mirrors now carry the tightest tolerances: a $\lambda/1000$ rms figure error on the two incoming primaries gives 2.4 % rms, and a 1 µrad tilt of both 3.7 %. (iii) The 0.1× telescope halves these numbers but doubles the intensity on the small mirrors. (iv) The Galilean form is 2 to 3.5 times more sensitive to figure errors than the Keplerian at the same magnification, though slightly less sensitive to primary tilt, because its design loss and its nearest mode are closer to the fundamental. In addition, the finesse barely matters. With a fixed $\lambda/300$ figure error the non-uniformity is 5.2 to 5.7 % at $G = 30$, 100, 300, 1000 and 3000 (finesse 47 to 7159), because the admixture is set by the mode spacing, not by the linewidth (Fig. 5).

The cavity filters input mismatch. A flat-top input 2% too narrow or too wide, or offset by 100 µm, changes the depth uniformity by less than 0.02 percentage points and the enhancement by at most 0.6 %. A Gaussian input with the optimal waist couples 0.918 of its power and changes

Table 5: Barrier-depth non-uniformity over the 10^6-site array (rms, median over realisations) and, in parentheses, the lowest enhancement relative to the analytic value. K: Keplerian, G: Galilean. Generated from the Fox–Li results by `scripts/cavity_report_tables.py`.

Perturbation	Plane folds, 150 mm	Plane folds, 300 mm	Telescope 0.2x K	Telescope 0.1x K	Telescope 0.2x G
G figure, rms, 0.001 waves	31 % (0.61)	17 % (0.84)	2.7 % (0.97)	1.3 % (0.98)	6 % (0.94)
G figure, rms, 0.003 waves	49 % (0.42)	44 % (0.41)	7.6 % (0.78)	3.9 % (0.82)	18 % (0.59)
G figure, rms, 0.01 waves	59 % (0.15)	57 % (0.19)	21 % (0.36)	14 % (0.21)	24 % (0.08)
P1 figure, rms, 0.001 waves	-	-	2.4 % (0.99)	1.3 % (0.96)	8.5 % (0.90)
P1 figure, rms, 0.003 waves	-	-	7.4 % (0.90)	3.9 % (0.73)	25 % (0.41)
P1 figure, rms, 0.01 waves	-	-	26 % (0.14)	14 % (0.09)	60 % (0.08)
G x^2 sag at beam half-width, 0.001 waves	5.7 % (1.00)	2.8 % (1.00)	0.82 % (1.00)	0.3 % (1.00)	0.87 % (1.00)
G x^2 sag at beam half-width, 0.003 waves	18 % (0.98)	8.7 % (1.00)	2.6 % (0.98)	0.65 % (0.99)	2.9 % (0.99)
G x^2 sag at beam half-width, 0.01 waves	61 % (0.86)	31 % (0.96)	12 % (0.70)	2 % (0.90)	5.5 % (0.78)
G tilt, 0.3 µrad	5.7 % (0.92)	3.3 % (0.97)	0.26 % (1.00)	0.23 % (1.00)	0.27 % (1.00)
G tilt, 1 µrad	15 % (0.68)	11 % (0.83)	0.43 % (1.00)	0.25 % (1.00)	0.53 % (1.00)
G tilt, 3 µrad	67 % (0.39)	26 % (0.58)	1.1 % (1.00)	0.38 % (1.00)	1.4 % (1.00)
G tilt, 10 µrad	80 % (0.26)	73 % (0.33)	3.7 % (0.97)	1 % (0.99)	5.1 % (0.96)
P1 tilt, 0.3 µrad	-	-	1.1 % (1.00)	0.6 % (1.00)	0.73 % (1.00)
P1 tilt, 1 µrad	-	-	3.7 % (0.97)	1.9 % (0.99)	2.4 % (0.99)
P1 tilt, 3 µrad	-	-	11 % (0.83)	5.8 % (0.93)	8.1 % (0.91)
P1 tilt, 10 µrad	-	-	33 % (0.56)	20 % (0.61)	24 % (0.59)

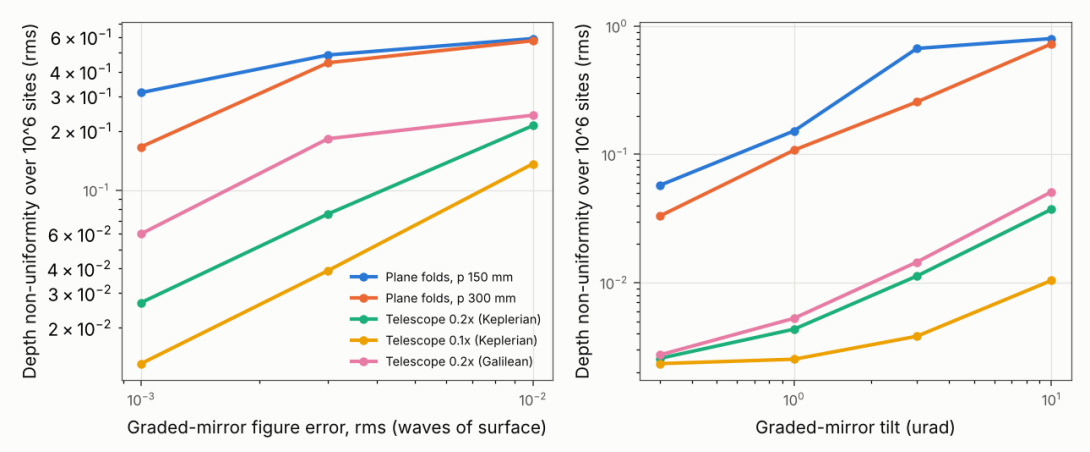


Figure 5: Barrier-depth non-uniformity over the 10^6-site array (rms over 101×101 sampled sites) versus the rms figure error of both graded mirrors (left, median over realisations) and the tilt of one graded mirror (right), for plane-fold rings of $p = 150$ and 300 mm and for telescope rings. Fox–Li strip model, 8192 points, $G = 300$, flat-top input, $w_z = 75$ µm.

the uniformity from 0.23 to 0.42 % rms. Thus the extracavity mode converter, the meta-optic role, needs only modest accuracy. Conversely, a ring without graded mirrors (plane folds and telescope, same flat-top input) gives 12 to 22 % rms at 0.86 to 0.87 of the enhancement, because the flat-top is not its eigenmode.

5.4 Can the graded mirror be a metasurface?

Quantising ϕ_G to 8, 16 or 32 levels gives 19, 12 and 4.5 % rms, at 0.14, 0.23 and 0.52 of the ideal enhancement. White pixel phase noise of 0.02, 0.05 and 0.1 rad gives 1.5, 3.8 and 7.4 % rms and reduces the enhancement to 0.89, 0.52 and 0.15 of the ideal. The SiN libraries of this project at 1075 nm (LPCVD and Si-rich, period 0.45 to 0.5 µm, height 1.3 to 1.6 µm) give a phase slope of 19 to 31 rad/µm over the 2π range. Hence 5 nm rms width noise is 0.10 to 0.15 rad in transmission, and about twice that in a reflective double pass (an estimate; no reflective library was simulated).[2] A metasurface graded mirror therefore needs sub-nanometre width control, or it must accept $G \approx 36$. The required phase is smooth (24 rad peak to valley, deflection at most 0.013), so a continuous freeform surface under an IBS coating is the natural realisation [REF?: freeform figuring at nm rms over mm apertures].

6 Correcting the residual non-uniformity

6.1 Method

The depth map of each axis is the product of the profiles of its two crossing beams, and each graded mirror shapes one of them, so the residual non-uniformity can be corrected by re-shaping the two graded mirrors. We put independent smooth figure errors, random tilts or both on every mirror of the ring: 8 telescope reflections and 2 graded mirrors for the 0.2× ring, and 4 folds and 2 graded mirrors for the plain 150 mm ring. Correlation lengths are referred to the crossing and scaled by the beam size on each mirror. We compare three corrections. (i) Open loop: G1 and G2 are redesigned through the half trips as measured,

$$\phi = \arg\Big(K_{\mathrm{out}}^{\dagger} T_{\mathrm{out}}\Big) - \arg(K_{\mathrm{in}} T_{\mathrm{in}})\,, \tag{5}$$

which reduces to Eq. (3) for a symmetric, unperturbed half trip. Here, K_{in} and K_{out} are the propagations to and from G and T_{in}, T_{out} the target fields at the crossing. Metrology noise is a smooth random error map of the stated surface rms on every mirror, plus 0.1 µrad of tilt error when tilts are present. (ii) Closed loop from the atoms, with no knowledge of the errors: each round the barrier depth at 101×101 sampled sites is measured with 0 or 1 % multiplicative noise per site and fitted as $a(\xi_A)\, b(\xi_B)$ by log-linear least squares with a smoothness penalty. Here, ξ_A and ξ_B are the transverse coordinates of the two beams. The target profiles are multiplied by $(a, b)^{-1/2}$ and G1, G2 redesigned with the nominal model, for six rounds. (iii) The same loop with a camera on the light transmitted by a cavity mirror instead of the atoms. Tables 6 and 7 give the results.

6.2 Open loop and the limits of mirror metrology

Perfect metrology removes 64 to 92 % of the non-uniformity of the telescope ring: $\lambda/1000$ errors on all ten mirrors fall from 2.0 to 4.8 % to 0.49 to 0.54 %, and 1 µrad tilts to 0.43 to 0.58 %. The floor arises because a phase error on a mirror away from G becomes partly an amplitude error at G and at the crossing, which a phase-only mirror cannot undo; in a test ring, a redesign through

[2] The noise model sets white noise per 2.9 µm grid cell, so it overstates the mode distortion from 0.5 µm pixels, whose scatter leaves the cavity at large angles. The loss, about σ^2 per bounce, does not depend on this.

Table 6: Telescope ring, 0.2×: depth rms over the 10^6-site array; two random realisations where shown, separated by a slash. Camera loop: 10 µm pixels, 48 bins. Generated by `scripts/cavity_report_tables.py`.

Errors on every mirror	Uncorrected	Perfect metrology	0.1 nm metrology	0.3 nm metrology	1 nm metrology	Atom loop, 6 rounds	Camera loop, 0.5 % noise	+ 1 % pixel response	+ 1 % flat field
λ/1000 figure, 2 mm	2 / 4.8 %	0.49 / 0.54 %	0.68 / 0.61 %	1.3 / 1.4 %	3.9 / 4.6 %	0.48 / 1.9 %	0.49 / 1.3 %	0.52 / 1.3 %	0.61 / 1.3 %
λ/1000 figure, 0.5 mm	3.6 %	2.4 %	2.3 %	2.3 %	3.9 %	2.3 %	2.8 %	2.8 %	3 %
λ/300 figure, 2 mm	7.1 / 22 %	1.4 / 1.7 %	1.5 / 2 %	1.9 / 2.9 %	4.1 / 7.2 %	1.5 / 48 %	1.6 / 6.7 %	1.6 / 6.8 %	1.6 / 6.9 %
tilts, 1 µrad rms	1.2 / 1.8 %	0.43 / 0.58 %	0.53 / 0.48 %	1.3 / 1.3 %	3.9 / 4.2 %	0.22 / 0.23 %	0.37 / 0.52 %	0.41 / 0.56 %	0.51 / 0.63 %
λ/1000 + 0.3 µrad tilts	2.9 %	0.6 %	0.71 %	1.4 %	3.9 %	0.54 %	0.61 %	0.64 %	0.72 %

Table 7: Plain ring with plane folds, $p = 150\,\mathrm{mm}$, as Table 6.

Errors on every mirror	Uncorrected	Perfect metrology	0.1 nm metrology	0.3 nm metrology	1 nm metrology	Atom loop, 6 rounds
λ/1000 figure, 2 mm	20 / 42 %	0.23 / 0.23 %	0.69 / 2.1 %	2.2 / 6.2 %	11 / 19 %	3.6 / 1.6 %
λ/1000 figure, 0.5 mm	24 %	0.26 %	0.7 %	2.2 %	11 %	0.94 %
λ/300 figure, 2 mm	42 / 1.3×10^2 %	0.24 / 0.25 %	0.73 / 2 %	2.2 / 6.2 %	11 / 19 %	49 / 37 %
tilts, 1 µrad rms	19 / 4.8 %	1.9 / 0.32 %	3.1 / 1.8 %	4.1 / 5.9 %	13 / 19 %	20 / 0.41 %
λ/1000 + 0.3 µrad tilts	56 %	1.9 %	3.1 %	4.1 %	13 %	44 %

a half trip with an error on its primary reduced the round-trip infidelity only fourfold, from 1.5×10^{-3} to 3.2×10^{-4}. Errors on a 0.5 mm scale leave 2.3 to 2.4 %.

The open loop is only as good as the metrology. At 0.1 nm rms per mirror the gain is kept, 0.3 nm leaves 1.3 to 1.4 %, and 1 nm (3.9 to 4.6 %) is no better than no correction. The best large-optics programmes reach 0.1 nm to 0.3 nm on the instrument. Advanced LIGO test-mass substrates were ion-beam figured to 0.08 to 0.23 nm rms over the central 160 mm (specification below 0.3 nm, power and astigmatism removed) [4, 5], and the metrology interferometer was specified for a repeatability below 0.15 nm rms, with 0.1 nm as the goal [6]. EUV projection mirrors were measured with 27 pm rms repeatability and 46 pm reproducibility, and one mirror reached 0.10 nm rms figure [7]. However, three differences make our requirement harder. First, it is absolute, per mirror, and must hold after coating; coating raised the LIGO end test mass from 0.18 to 0.69 nm rms and the input test mass from 0.15 to 0.31 nm [4], more than our whole tolerance. Repeatability is not absolute accuracy, and independent absolute calibrations of reference flats agree only to `[VALUE?: approx. 0.4 nm rms, verify]`. Second, our mirrors are off-axis cylindrical or freeform surfaces used at 15° to 38° incidence, each needing its own null test. Third, the map must describe the mirror as mounted, in vacuum and at the circulating power, and must be registered to the beam to a fraction of the error correlation length (below 0.1 mm on the small mirrors). Thus such metrology belongs to fabrication, to bring each mirror to $\lambda/1000$, and not to in-situ correction, which must be closed-loop.

6.3 Closed loop from the atoms and from cameras

The closed loop needs no mirror metrology. Of its eight runs on the telescope ring, six converge within four to five rounds to the perfect-metrology level or below: 0.48 % for $\lambda/1000$, 0.22 and 0.23 % for tilts (the design floor), 0.54 % for the combined case, 2.3 % for 0.5 mm errors and 1.5 % for the first $\lambda/300$ realisation. One $\lambda/1000$ realisation stalls at 1.9 % with 0.94 of the enhancement. The $\lambda/300$ realisation that starts at 22 % (0.48 of the enhancement) diverges. Thus the feedback works when the uncorrected mode is close to the flat-top (starts up to about 7 % here) and fails when the mode is no longer the designed one. Per-site measurement noise of 1 % changes nothing, because the separable fit averages over 10^4 sampled sites.

A lower loop gain does not help. At gain 0.25 for 12 rounds the stalled $\lambda/1000$ realisation ends at 1.89 % (1.93 % at gain 0.5), and the 0.5 mm case at 2.1 to 2.3 % with its enhancement falling to 0.85. From the 22 % start, gains of 0.25 and 0.1 still collapse the enhancement, to 0.02 and 0.07, although at gain 0.1 the depth rms falls to 12 %. Hence the stall and the divergence are properties of the update rule, which redesigns with the nominal model and a phase-only mirror, not of its gain. A model-based update, in which the measured depths are used to estimate the mirror errors, is the next candidate; we have not simulated it.

The light transmitted by a cavity mirror carries the circulating profile of each beam, and a camera behind an imaging relay conjugate to the crossing (assumed ideal here) sees $|E_A|^2$ and $|E_B|^2$ directly. With 0.5 % fresh noise per pixel, the same update rule converges as well as the atom loop and more reliably: 0.49 and 1.3 % for $\lambda/1000$ errors (atom loop 0.48 and 1.9 %), and 6.7 % from the 22 % $\lambda/300$ start that made the atom loop diverge. It is slightly worse for tilts (0.37 to 0.52 %, against 0.22 to 0.23 %), because it sees only the crossing plane and not the propagation of each beam across the array. A 1 % static per-pixel response error changes nothing, because the bins average it. A 1 % rms smooth flat-field error, which the loop cannot distinguish from a real profile error, adds at most 0.14 percentage points. A 50 ppm mirror leaks about 5 mW of a 100 W circulating beam, ample for a camera. Thus a camera loop can run continuously against drift, with the atoms as the occasional absolute calibration of the camera's flat field and of the relay.

The required stroke is 1 to 7 nm rms of graded-mirror surface in all converged cases. The correction could therefore be written by iterative refiguring, or applied in situ by a deformable or

thermally actuated graded mirror; the loss and stability of an intracavity deformable mirror at kW/cm^2 intensities are open [VALUE?]. The plain ring is fully correctable with perfect metrology (0.23 to 0.26 %) but not in practice: metrology of 0.1 nm leaves 0.7 to 2.1 %, and the closed loop converges in only four of nine error sets (from starts of 4.8 to 42 %). Correction does not replace the telescope.

6.4 Non-separable errors

A two-dimensional Fox–Li model of the 0.2× ring (1024 × 192 points over 14 × 2.4 mm, vertical focal length 30 mm on the primaries, two-dimensional design, $G = 30$ for speed) tests whether non-separable errors spoil the correction. Isotropic smooth $\lambda/1000$ errors on all ten mirrors vary across the vertical beam on the small mirrors, so they are non-separable (Table 8).

Table 8: Two-dimensional check of the 0.2× ring with non-separable $\lambda/1000$ errors.

Case (2D)	Depth rms	Enhancement / analytic
Ideal 2D design	0.23 %	1.000
$\lambda/1000$ on all mirrors, uncorrected	2.2 %	0.998
Perfect metrology, full 2D phase on G	1.6 %	0.999
Perfect metrology, correction projected onto x only	0.53 %	0.999
Closed loop from the atoms, x-only updates, 4 rounds	0.44 %	0.998

The ideal two-dimensional design reproduces the strip model (0.232 against 0.231 %), and the vertical mode on G (0.648 mm) reproduces the ABCD design (0.65 mm). Non-separable errors do not prevent the correction: an x-only, i.e. cylindrical, correction recovers most of it, and the closed loop reaches 0.44 % in four rounds, still falling. The full two-dimensional phase-only redesign does worse than its own x projection. We attribute this to the phase-only matching of the whole field, which spreads the residual amplitude error over the vertical profile, including the atom plane; we did not test this further.

6.5 The vertical ring

The vertical ring is modelled as the product of two one-dimensional modes, each driven and locked separately: the horizontal direction with the telescope and a 6.9 mm equivalent width, and the vertical order-10 flat-top of 0.19 mm flat half-height. In the adopted fold geometry the horizontal fundamental loses 3.6×10^{-7} per round trip and its nearest competing mode lies 0.30 rad away with 7.8×10^{-3} loss, against 0.14 rad and 3.1×10^{-4} for an unconstrained 0.2× design. The ideal depth is uniform to 0.19 % rms (minimum over mean 0.988). $\lambda/1000$ errors on all ten mirrors in the horizontal direction give 5.7 and 12 % uncorrected (against 3.6 and 36 % for the unconstrained design) and 0.19 and 0.20 % after a perfect-metrology redesign; in the vertical direction they give 1.5 and 0.7 %, and 0.33 % after redesign. Atom feedback, with errors in both directions at once and the depth map fitted as a product of a function of the horizontal transverse coordinate and a function of the position along the diagonal, reaches 0.17 and 0.19 % in six rounds from 5.9 and 12 %. For this ring the fit is exact, so the loop converges to the design value. The lattice planes stay flat to at most 0.5 µm peak to valley, inside the ±1.7 µm depth of focus of the imaging objective. The narrow vertical flat-top is the weaker direction for power: its design loss rises to 2.9×10^{-5} on the longer path, and the vertical errors reduce its enhancement to 0.80 to 0.90 of the ideal even after correction, i.e. 10 to 25 % more input power for this ring.

7 Intensities, defects and heating on the mirrors

7.1 In-plane rings

The peak intensity of each beam at the atoms is I_1 of Eq. (4). On a mirror it is I_1 times the one-dimensional peak ratio from the Fox–Li model (5.7 on P2 and 6.5 on G for the 0.2× design) times the vertical size ratio. Our defect criterion is that a fully absorbing disc of radius a on a half-space heats by Ia/κ_{th}, limited to 50 K. Here, κ_{th} is the thermal conductivity of the substrate. Table 9 gives the result.

Table 9: Peak intensities and the largest tolerable absorbing defect, in-plane rings, 0.2 mK, 10^6 sites.

Surface	Peak intensity	Largest absorbing defect, fused silica / sapphire
Cell wall at 15 mm [CONFIRM?] (through-cell variant)	14.6 kW/cm^2	0.47 / 11 µm
Telescope primary P1	4.5 kW/cm^2	1.5 / 37 µm
Telescope secondary P2	19.3 kW/cm^2	0.36 / 8.6 µm
Graded mirror G	17.4 kW/cm^2	0.40 / 9.5 µm

At equal radial frequency (0.86 mK) every intensity is 4.3 times higher and every defect limit 4.3 times smaller (0.08 to 0.35 µm on fused silica). For comparison, the present 18.4 mm metasurface at 200 W sees 75 W/cm^2 and tolerates 92 µm. The cavity therefore trades a kilowatt laser and a hot metasurface for watt-level input and mirrors at kW/cm^2. Point absorbers at these intensities limit the stored power of Advanced LIGO [8]. The fully absorbing disc model is pessimistic for sub-micron particles, whose absorption cross-section is below the geometric one.

7.2 Vertical ring

The adopted fold geometry of the vertical ring compresses the 6.9 mm horizontal width on P2 and G with its 0.15× telescope, while the 0.19 mm vertical flat-top has no telescope and spreads only by diffraction. We compute the peak intensity of the beam cross-section on each mirror from the two one-dimensional Fox–Li modes, with the 785 W that circulate in this ring at 0.2 mK. We compare the adopted design with two variants that add a Keplerian telescope in the vertical direction at the same mirror positions, with magnification 4 or 8 and $f_{1z} + f_{2z} = 184$ mm, i.e. P1 curved in the plane of incidence with $f_{1z} = 36.8$ or 20.4 mm (Table 10).

Table 10: Vertical ring at 0.2 mK: peak intensity of the beam cross-section on P1, P2, G and the return P2, P1 (kW/cm^2); smallest tolerable defect; nearest vertical mode and fundamental loss; $\lambda/1000$ vertical errors, two realisations, uncorrected and after a perfect-metrology redesign, with G/G_0 in parentheses.

Variant	P1 / P2 / G / P2′ / P1′	Smallest defect, silica / sapphire	Nearest mode; fundamental loss	Errors: uncorrected; redesign
Adopted, no vertical telescope	32 / 170 / 145 / 112 / 14	0.041 / 0.97 µm (P2)	2.15 rad, 4.0×10^{-3}; 2.9×10^{-5}	1.5 and 0.69% (0.58, 0.97); 0.33 and 0.33% (0.93, 0.96)
Vertical telescope 4×	32 / 51 / 55 / 55 / 35	0.13 / 3.0 µm (G)	1.67 rad, 1.1×10^{-4}; 6×10^{-8}	5.0 and 8.3% (0.99, 0.98); 2.0 and 1.4% (1.00, 1.00)
Vertical telescope 8×	32 / 27 / 32 / 32 / 37	0.19 / 4.4 µm (P1′)	2.65 rad, 4.8×10^{-4}; 5.3×10^{-6}	23 and 6.8% (0.79, 0.98); 2.8 and 0.41% (0.99, 1.00)

The ideal depth uniformity is 0.19, 0.073 and 0.16 % rms. As folded, the vertical ring puts up to 170 kW/cm^2 on P2, nine times the in-plane maximum, and tolerates absorbing defects only

up to 0.04 µm on fused silica.[3] An 8× vertical telescope brings every mirror to 27 to 37 kW/cm^2, i.e. 1.4 to 1.9 times the intensity at the atoms, which cannot be lowered on P1 without moving it further from the atoms. The surface intensity is lower than the cross-section value by cos AOI (0.90 on P2 and 0.85 on G at the smallest angles of the layout). The vertical telescope also lowers the design loss of the narrow mode and keeps its enhancement at 0.99 to 1.00 after a perfect-metrology redesign (0.93 to 0.96 without it). However, it makes the vertical mode less discriminating (1.1×10^{-4} loss for 4×) and the uncorrected depth more sensitive to figure error (5 to 23 % against 0.7 to 1.5 %). Our choice for the vertical ring is therefore a compromise. First, sapphire substrates on all its mirrors, which raise the defect limit 23 times. Second, a vertical telescope of 4× to 8×, with closed-loop correction of the vertical profile as a requirement; the closed loop and the layout with the added vertical curvature are not simulated. At equal radial frequency this ring circulates 1059 W instead of 785 W, and every intensity scales by 1.35.

7.3 Thermal deformation of the graded mirror

Table 11 gives the absorption-induced bulge of a graded mirror of an in-plane ring (102 W circulating; equal-area disc of 0.43 mm radius on a 25.4 mm diameter, 6.35 mm thick fused-silica or sapphire mirror; surface bulge $(1 + \nu)\,\alpha \int \Delta T \,\mathrm{d}z$). Here, ν is the Poisson ratio and α the expansion coefficient of the substrate.

Table 11: Thermal deformation of the graded mirror of an in-plane ring.

Absorber on G	Absorbed power	Surface ΔT	Bulge, centre to edge	Bulge rms over the spot
IBS HR, 1 ppm [VALUE?]	0.1 mW	0.05 K	0.004 nm	10^{-6} waves
IBS HR, 10 ppm	1 mW	0.5 K	0.035 nm	10^{-5} waves
LPCVD SiN metasurface (3.5×10^{-6})	0.36 mW	0.19 K	0.012 nm	3×10^{-6} waves
Si-rich SiN metasurface (5.9×10^{-4}), fused silica	60 mW	32 K	2.1 nm	5.6×10^{-4} waves
Si-rich SiN metasurface (5.9×10^{-4}), sapphire	60 mW	1.3 K	0.9 nm	2.4×10^{-4} waves

With IBS coatings the deformation is three orders of magnitude below the $\lambda/1000$ figure tolerance, while a Si-rich SiN film inside the cavity would reach the tolerance on its own. We fed the bulge into the Fox–Li ring with a profile proportional to the intensity on G, which is sharper than the thermal profile and hence conservative, on both graded mirrors of the 70 mm ring. The depth rms changes from 0.231 to 0.232 % for 10 ppm IBS and to 0.252 % for 100 ppm; the Si-rich SiN film raises it to 0.53 % on fused silica and 0.32 % on sapphire, and the enhancement changes by less than 4×10^{-4}. A static bulge is also within reach of the closed loop. The vertical ring's mirrors absorb 7.7 times more on smaller spots; we did not compute their deformation, and scaling the centre bulge with the absorbed power gives about 0.3 nm for 10 ppm, a rough estimate. The cell windows of the through-cell variant absorb 4 ppm per pass in bulk (0.4 mW at 102 W), which is negligible thermally, but their surfaces carry the defect constraint of Table 9; this is the reason to prefer mirrors in vacuum.

8 Site quality, optical access and loading

8.1 Trap frequencies and registration

Table 12 gives the trap frequencies of the lattice (plane-wave model, numerical Hessian, equal to the analytic $(\pi/a)\sqrt{2U/m}$ to 10^{-6}).

[3] An earlier estimate in the working report assumed a threefold beam expansion on the vertical ring's mirrors (2.2 kW/cm^2); the adopted geometry does not provide it.

Table 12: Trap frequencies at the lattice site and of a tweezer of the same depth.

Barrier per axis	In-plane	Vertical (8 µm)	Tweezer, radial / axial
0.2 mK	19.8 kHz	9.9 kHz	41.1 / 11.5 kHz
0.3 mK	24.2 kHz	12.1 kHz	50.4 / 14.1 kHz
0.86 mK	41.0 kHz	20.5 kHz	85.3 / 23.8 kHz

At equal depth the lattice site has 0.48 of the tweezer's radial frequency, because its $\cos^2$ well spans the full pitch, so equal radial frequency costs a factor 4.3 in depth and power. A cost that the power table hides is the light shift at the site: each of the three incoherent axes reaches $4I_1$ there, twice its mean, so the atoms sit at $3U$ of light shift (0.6 mK for a 0.2 mK barrier), against U in a tweezer. Differential light shifts and their noise are thus three times larger at equal barrier depth.

A differential wavefront error $\mathrm{d}w$ between the two beams of an axis moves a site by $\mathrm{d}w/(2\sin\theta)$, i.e. 3.7 nm per nanometre. A crossing-angle change of 1 µrad changes the pitch by $\cot\theta \times 1$ µrad = 7.4 ppm and moves the edge sites by 15 nm, and a wavelength change of 1 ppm moves them by 2 nm. A change of the optical path between the two crossings of one ring shifts that lattice rigidly by the same 3.7 nm per nanometre; the shift is common to all sites and can be tracked, but it sets the path-stability requirement between the crossings (Section 9). In the tolerance cases the non-affine site displacement is at most 37 nm rms for every telescope case with depth non-uniformity below 5 %. Thus site positions are stable to a small fraction of the 4 µm pitch, provided the crossing angle is held to about 1 µrad.

8.2 Layout

With the primary 70 mm from the crossing and the same fold angle (157°) for both rings, the two in-plane rings fit within ±150 mm. Their mirrors and beams keep at least 15.5 mm from each other (mirror apertures and a 3 mm beam allowance included), the folded beams stay at least 35.8 mm from the atoms, and both array diagonals stay clear by 8.2 mm beyond a 40 mm radius, the corridor left for the vertical ring. The vertical ring occupies the vertical plane on one diagonal, below the atom plane, and keeps 37.8 mm from the in-plane rings' mirrors. Mounts, coils and the vacuum envelope are not designed.

8.3 Optical access for the MOT, imaging, pumping and Raman beams

For every line through the atoms we compute the clearance of a beam of given radius (250 mm each way) from all twenty mirrors of the three rings, each with a 5 mm mount allowance, and from the imaging objective. The objective is the 42 mm aplanatic doublet that images 10^6 sites, with its first plate 38 mm above the atoms and a 26 mm radius including its mount `[CONFIRM?]`. Ring beams are light only and may be crossed; windows are assumed wherever needed `[CONFIRM?: chamber]`. Table 13 and Fig. 6 give the result.

The free band of directions lies between about 43° and 75° from vertical at all azimuths, except near the vertical ring's plane: the objective closes the cone above, and the ring mirrors close the horizontal. A six-beam MOT with its three axes along the body diagonals of a cube standing on a corner, each about 55° from vertical, fits with margin and leaves room for pumping and Raman beams. Global qubit rotations can also use microwaves (9.2 GHz, no optical access), with single-site addressing through the objective.

Table 13: Optical access around the three rings and the objective.

Beam set	Result
Standard MOT, axes x, y, z (20 mm beams)	blocked: horizontal beams hit in-plane mirrors (−17.5 mm), the vertical beam hits the objective (−36 mm)
Standard MOT on the diagonals and z	blocked (−14 mm, −6.8 mm, −36 mm)
Orthogonal MOT triad, axes 54° to 55° from vertical (azimuths 15°, 75°, 135°)	free, clearance at least 15.9 mm for 20 mm beams
Lines free for 20 / 10 mm beams	34 / 45 % of all lines through the atoms
Four further lines for optical pumping and Raman beams (56.5° from vertical; azimuths 117°, 155°, 297°, 333°), at least 15° from the MOT axes	clearance 26.9 mm for 10 mm beams, 21.9 mm even for 20 mm beams
Imaging and addressing	the objective's own port above the atoms (NA 0.5; the free band starts at 43° from vertical)

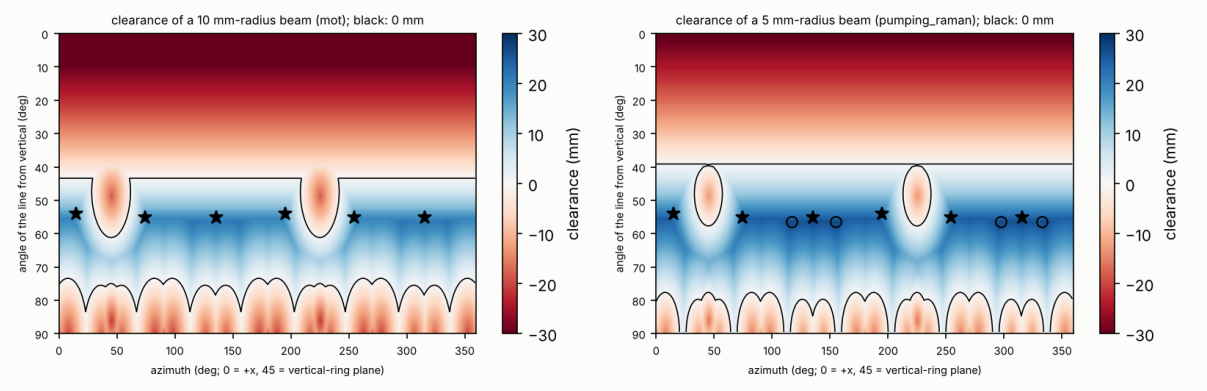


Figure 6: Clearance of a beam through the atoms versus its direction (angle of the line from the vertical; azimuth from $+x$; the vertical ring's plane is at 45° and 225°), for 20 mm (left) and 10 mm (right) beams; black contour: zero clearance. Stars: the axes of the tilted six-beam MOT and their opposite ends; circles: the four pumping and Raman lines. Obstacles: all ring mirrors with a 5 mm mount allowance and the 42 mm objective from 38 mm above the atoms [CONFIRM?].

8.4 How many atoms loading needs

A lattice site is $4 \times 4 \times 8\,\mu\text{m}^3 = 1.3 \times 10^{-10}\,\text{cm}^3$. With the usual collisional loading of about one atom in two sites [VALUE?: about 0.5 filling, light-assisted collisions], each site of the loaded plane must first see about two atoms, i.e. a density of about $1.6 \times 10^{10}/\text{cm}^3$ across the 4 mm array. A MOT cloud that keeps this density at the array corners (2.83 mm) with a peak three to five times higher holds about 5×10^9 atoms, because most of its atoms lie in other planes. Accumulating atoms in the in-plane sheets first, which span about 19 of the 8 µm planes, reduces the need to about 4×10^7 atoms delivered into the sheets, i.e. a MOT of 10^8 to 10^9 atoms of at least 6 mm extent. Demonstrated Cs numbers are $1.1(1) \times 10^9$ atoms captured in an optical molasses from a source of about 10^{10} atoms per second [9], and two-dimensional MOT fluxes of 4.02×10^{10} atoms per second [10]. A tilted six-beam MOT loaded from a two-dimensional MOT can therefore supply the atoms.

A single-beam grating MOT fits geometrically: a 22 mm chip 8 mm below the atoms clears every ring beam by at least 6.9 mm, its three 45° diffracted beams clear all mirrors and the objective by 6.7 mm [VALUE?: diffraction angle, set by the grating period], and the vertical axis below it is free for 96 mm (to G2) for a retro-reflector. Its single input beam must come from above, along the objective axis. However, demonstrated grating MOTs are small for this purpose: $2.46(7) \times 10^8$ Rb atoms when loaded from a two-dimensional grating MOT [11], and for Cs only 7.0×10^6 atoms, and only with an additional retro-reflected beam of 69 % of the input intensity; at 24 % no atoms were trapped [12]. A Cs grating MOT is therefore one to two orders of magnitude short of a single-shot load of 10^6 sites, and would need iterative accumulation with the stored atoms shielded from the MOT light. We therefore recommend the tilted six-beam MOT, with the grating MOT as a compact fallback for smaller arrays or with iterative loading.

9 Noise and heating requirements

We use the parametric-heating rates of Savard, O'Hara and Thomas [13]. Intensity noise gives an energy e-folding time T_I and position noise an energy-doubling time T_x,

$$\frac{1}{T_I} = \pi^2 \nu^2 S_\epsilon(2\nu), \qquad \frac{1}{T_x} = \frac{\pi^2 \nu^2 S_x(\nu)}{\langle x^2 \rangle}, \qquad \langle x^2 \rangle = \frac{k_B T}{m(2\pi\nu)^2}. \tag{6}$$

Here, ν is the trap frequency in hertz, S_ϵ and S_x are one-sided power spectral densities of the fractional intensity and of the trap position, m is the atomic mass and T the temperature. Our implementation reproduces their numerical examples ($3.2 \times 10^{-6}/\sqrt{\text{Hz}}$ and $3.2 \times 10^{-12}\,\text{m}/\sqrt{\text{Hz}}$ for 100 s at 10 kHz). The position term is a constant heating rate, $\dot{E} = 4\pi^4 m \nu^4 S_x$, independent of temperature. We quote it for atoms at $T = 10\,\mu\text{K}$ [VALUE?: temperature after cooling in the lattice], so that a 10 s doubling time is 1 µK/s, about one vibrational quantum per second at 20 kHz.

The site shift of the self-intersecting ring is $a\,\mathrm{d}\phi/(2\pi)$, where $\mathrm{d}\phi$ is the change of the phase accumulated between the two passes through the crossing, over the path $L_1 = L_{\text{rt}}/2$. This gives four couplings. (i) A normal displacement d of one ring mirror at incidence AOI, with the laser locked to the round trip L_{rt}, shifts the sites by $a\,(2d \cos \text{AOI}/\lambda)(1 - L_1/L_{\text{rt}})$, i.e. by $3.2\,d$ for the in-plane rings and $6.4\,d$ for the vertical ring at 30°. (ii) A laser frequency change $\mathrm{d}\nu$ that the lock does not remove shifts them by $aL_1\,\mathrm{d}\nu/c$ (7.7×10^{-15} and 2.2×10^{-14} m/Hz). (iii) A change $\mathrm{d}\theta$ of the crossing half-angle changes the pitch by $a \cot\theta\,\mathrm{d}\theta$ and moves the edge sites by $(S/2)\cot\theta\,\mathrm{d}\theta$ (14.7 and 29.7 mm/rad). (iv) A residual laser-cavity detuning $\mathrm{d}\nu_r$ modulates the circulating intensity by $-(2\,\mathrm{d}\nu_r/\kappa_c)^2$, where κ_c is the full linewidth. For Gaussian $\mathrm{d}\nu_r$ with a flat one-sided spectrum S_ν over a bandwidth $B = 100$ kHz this gives

$$S_\epsilon \approx \frac{32 S_\nu^2 B}{\kappa_c^4}, \qquad S_\epsilon = \frac{64\,\delta_0^2 S_\nu}{\kappa_c^4} \text{ (linear term, static offset } \delta_0). \tag{7}$$

The linewidths are those of the over-coupled design point with the 258 and 181 MHz free spectral ranges. Table 14 gives the requirements.

Table 14: Noise requirements as one-sided amplitude spectral densities: position, displacement, frequency and angle at ν; intensity at 2ν; lock residual flat over 100 kHz, zero offset. Atoms at 10 µK; one mirror at 30° incidence.

Lattice, depth	ν (kHz)	Heating time	RIN at 2ν (dBc/Hz)	Site position (pm/$\sqrt{\text{Hz}}$)	One mirror (pm/$\sqrt{\text{Hz}}$)	Laser frequency (Hz/$\sqrt{\text{Hz}}$)	Crossing angle, edge (nrad/$\sqrt{\text{Hz}}$)	Lock residual $G = 300$ / 1500 (Hz/$\sqrt{\text{Hz}}$)
In-plane, 0.2 mK	19.8	10 s	−106	1.0	0.32	132	0.070	22 / 4.4 (7.0 / 1.4 kHz rms)
In-plane, 0.2 mK	19.8	100 s	−116	0.32	0.10	42	0.022	12 / 2.5
In-plane, 0.86 mK	41.0	10 s	−112	0.24	0.074	31	0.016	15 / 3.0
Vertical, 0.2 mK	9.9	10 s	−100	4.1	0.64	186	0.14	22 / 4.3
Vertical, 0.86 mK	20.5	10 s	−106	0.95	0.15	43	0.032	15 / 3.0

Seven conclusions follow. (i) The cavity passes the laser's relative intensity noise unchanged below $\kappa_c/2$ (206 kHz at $G = 300$), so the input beam itself needs −106 dBc/Hz at 40 kHz for 10 s at 0.2 mK, and −116 dBc/Hz for 100 s. This is a requirement on the amplifier and the intensity servo, not on the cavity `[VALUE?: RIN of a stabilised 1075 nm fibre amplifier at 20 to 80 kHz]`. (ii) The residual detuning must stay below 22 Hz/$\sqrt{\text{Hz}}$ over 100 kHz (7.0 kHz rms, 1.7 % of the 413 kHz linewidth) at $G = 300$, and below 4.4 Hz/$\sqrt{\text{Hz}}$ (1.4 kHz rms) at $G = 1500$. The requirement scales as κ_c^2 in the quadratic regime, so $G = 1500$ is five times harder; a static offset of 1 % of the linewidth makes the linear conversion comparable (26 and 5.2 Hz/$\sqrt{\text{Hz}}$). This quantifies the reason for working at $G \approx 300$. (iii) Position noise is the stricter condition. A site motion of 1 pm/$\sqrt{\text{Hz}}$ at 20 kHz, one quantum per second, corresponds to a single-mirror displacement of 0.32 pm/$\sqrt{\text{Hz}}$, or 0.1 pm/$\sqrt{\text{Hz}}$ for 100 s. Coating Brownian noise is far below this `[VALUE?: approx. 1e-17 m/rtHz at kHz, verify]`; the relevant source is the excitation of mount and spacer resonances in the 10 to 40 kHz band by acoustic noise, pumps and the lock actuator. Hence the ring should be a monolithic, low-expansion assembly, as in Ref. [1], with its lowest mechanical resonances either above about 50 kHz or well damped `[VALUE?: mount vibration spectrum]`. (iv) With the laser locked to the ring, its frequency follows the round-trip length, and coupling (i) already includes this through the factor $(1 - L_1/L_{\text{rt}})$; the laser-frequency column applies to noise that the lock does not remove, i.e. above the lock bandwidth and below $\kappa_c/2$, and a lock bandwidth above 2ν avoids it. (v) Edge sites move by 14.7 mm per radian of crossing-angle change, so 0.07 nrad/$\sqrt{\text{Hz}}$ is needed at 20 kHz; for a mirror mount with a 10 mm lever this is 0.7 pm of differential motion, comparable to (iii) and not a separate technology. (vi) The cavity is a mode filter for input pointing jitter. The nearest transverse mode of the 70 mm ring lies 0.216 rad from the fundamental per round trip, i.e. 8.9 MHz, against a 206 kHz half-linewidth at $G = 300$; input misalignment therefore couples into it with about 0.02 of the amplitude it would have without the cavity, and the site positions are set by the mirrors, not by the input beam. (vii) The vertical lattice, with its lower trap frequency at 0.2 mK, is the least sensitive to intensity noise (−100 dBc/Hz at 20 kHz), but its larger lever gives a similar mirror requirement (0.64 pm/$\sqrt{\text{Hz}}$).

These are requirements, not predictions: we have not simulated the time-dependent depth and site position, nor estimated the vibration spectrum of a real mount.

10 Discussion and conclusions

What is shown, within the one-dimensional scalar model and the parameters explored, is the following. A cavity cannot give a static meta-optic tweezer array a tenfold power reduction ($G \leq 3.2$). A three-ring lattice with graded-phase mirrors and intracavity telescopes supplies 10^6 Cs sites at 0.2 mK with 3.5 W of input at $G = 300$, a factor 300 below the same lattice in free space and 400 below hologram tweezers, with the depth uniform to 0.23 % rms by design. Mirror errors of $\lambda/1000$ give a few per cent, and closed-loop re-shaping of the graded mirrors, from the atoms or from cameras on the transmitted light, removes most of it with 1 to 7 nm of stroke.

What is model-dependent is the following. Every Fox–Li result treats the horizontal and vertical profiles as separable, except one two-dimensional check per question, and uses static perturbations with one or two random realisations per case. The vertical ring is modelled as the product of two one-dimensional modes. The noise analysis gives requirements from analytic heating rates, not a time-dependent simulation. The layout is a first-order design without mounts, coils or the vacuum envelope.

What is extrapolated is the following. The intensities of 4.5 to 19 kW/cm^2 in the in-plane rings, and 27 to 55 kW/cm^2 in the vertical ring with a vertical telescope (up to 170 kW/cm^2 without it), are comparable to those on the test masses of gravitational-wave detectors, where point absorbers limit the stored power [8]. Their consequence for a cold-atom apparatus, in which the mirrors sit close to alkali vapour, has not been tested.

The main limitation is therefore no longer laser power or metasurface heating but the mirrors: figure and alignment stability at the $\lambda/1000$ and 1 µrad level between corrections, absorbing defects no larger than 0.36 µm on the fused-silica mirrors of the in-plane rings and 3 to 4.4 µm on sapphire mirrors of the vertical ring with its telescope, and vibration below 0.3 pm/$\sqrt{\text{Hz}}$ at the trap frequency. The next steps are (i) a model-based closed loop for large starting errors, and the closed loop and layout of the vertical ring with a vertical telescope; (ii) a time-dependent simulation of depth and site position under laser and lock noise, with a vibration model of the ring mounts; and (iii) the absorption-induced deformation of the vertical ring's mirrors, and the loading of a single plane of the vertical lattice from the tilted MOT.

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A Design summary

Table 15: Parameters of the adopted design for 10^6 sites at a 0.2 mK barrier per axis (1075 nm, Cs). Input powers per ring are $P_{\mathrm{circ}}/(0.95\,G)$.

Quantity	Each in-plane ring (two)	Vertical ring
Lattice period, crossing angle 2θ	4 µm, 15.4°	8 µm, 7.7°
Mode at the atoms	order-20 flat-top, 5.51 mm equivalent width, times Gaussian sheet $w_z = 75$ µm	flat-top 6.9 mm wide, times order-10 flat-top of 0.19 mm half-height
Telescope per half trip	0.2× Keplerian, $f_1 = 100$ mm, P1 at 70 mm, $L = 100$ mm	0.15×, $f_1 = 160$ mm, P1 at 110 mm, $L = 120$ mm
Round trip, free spectral range, reflections	1.16 m, 258 MHz, 10	1.66 m, 181 MHz, 10
Circulating power	102 W	785 W
Input power at $G = 300$ (mode matching 0.95)	0.36 W	2.75 W
Linewidth (FWHM) at $G = 300$ / 1500	413 / 82 kHz	289 / 58 kHz
Peak intensity on the mirrors	4.5 (P1), 19.3 (P2), 17.4 (G) kW/cm^2	14 to 170 kW/cm^2 as folded; 27 to 37 kW/cm^2 with an 8× vertical telescope
Trap frequency along the lattice axis	19.8 kHz	9.9 kHz
Depth uniformity: design / $\lambda/1000$ errors / after atom feedback	0.23 / 2.0 to 4.0 / 0.42 to 0.75 %	0.19 / 5.7 to 12 / 0.17 to 0.19 %
Static figure and tilt before correction	$\lambda/1000$ rms, below 1 µrad	$\lambda/1000$ rms
Noise for a 10 s heating time at 10 µK	RIN below −106 dBc/Hz at 40 kHz; 0.32 pm/$\sqrt{\mathrm{Hz}}$ per mirror at 20 kHz	RIN below −100 dBc/Hz at 20 kHz; 0.64 pm/$\sqrt{\mathrm{Hz}}$ per mirror at 10 kHz

The total circulating power is 989 W and the total input 3.5 W at $G = 300$ (0.69 W at $G = 1500$), against 1.04 kW for the same lattice in free space and 1.40 kW for meta-hologram tweezers.

B Reproducibility

All results come from the scripts and configurations of the repository; each run writes a `params.json` with its configuration, git state and package versions next to its results. Table 16 lists them. Every script is run as `python scripts/<name>.py configs/<name>.yaml`, except the table generator.

Table 16: Scripts, configurations and results behind each section.

Section	Script and configuration	Results
2, 4	`cavity_budget`, `cavity_tweezer_bounds`	`results/cavity_budget/budget.json`, `results/cavity_tweezer_bounds/bounds.json`
3, 8	`cavity_layout`, `cavity_vertical_layout`, `plot_cavity_sketch`, `cavity_lattice`	`results/cavity_layout/`, `results/cavity_vertical_layout/`, `results/cavity_lattice/lattice.json`
5	`cavity_foxli`; tables by `cavity_report_tables.py latex docs/report/tables`	`results/cavity_foxli/foxli.json`
6	`cavity_correction`, `cavity_2d_check`, `cavity_vertical` (`cavity_vertical_chosen.yaml`)	`results/cavity_correction/`, `results/cavity_2d_check/`, `results/cavity_vertical_chosen/`
7	`cavity_thermal`, `cavity_vertical_mirrors`, `cavity_thermal_foxli`	`results/cavity_thermal/`, `results/cavity_vertical_mirrors/`, `results/cavity_thermal_foxli/`
8 (access)	`cavity_access`	`results/cavity_access/access.json`, `access.png`
9	`cavity_noise`	`results/cavity_noise/noise.json`

The core model is in `src/metatrap/cavity/` (`budget.py`, `lattice.py`, `foxli.py`) and is tested by `tests/test_cavity.py`.

C Provenance and open placeholders

Every number in this report is of one of four kinds. SIM values come from simulations executed for this report and hold only within the simulated regime; they should be reviewed before submission. DERIVED values are calculations from stated inputs, and SOURCE values are taken from the references listed above, each of which was retrieved and read. The precedent values of Park et al., Norcia et al., LIGO, Zeiss, the Cs MOT and grating-MOT experiments, and the heating rates of Savard et al. are SOURCE; the cesium polarisability is the repository's implementation of Steck's data (DERIVED from a source not re-read for this report); the tweezer anchors of 143 and 80 mK site per watt are SIM from earlier studies of this project; all Fox–Li, correction, two-dimensional, vertical-ring, deformation and access results are SIM; the budget, layout, trap frequencies, loading estimate and noise requirements are DERIVED. The full ledger, with one row per value and its file pointer, is in the working edition (`docs/cavity_concept.md`, Verification notes).

The open placeholders are: AR loss per surface (2.5×10^{-4} to 10^{-3}); Brewster residual loss (10^{-4}); IBS absorption (1 ppm); refractive objective (0.95) and ideal relay (0.995 per pass); mode-converter coupling (0.95); cell wall distance (15 mm) and cell size; freeform figuring at nanometre rms over millimetre apertures (reference); absolute flat-calibration agreement (about 0.4 nm); loss and stability of an intracavity deformable mirror; objective position and mount, chamber windows and mounts; grating diffraction angle; collisional filling (about 0.5); atom temperature in the lattice (10 µK); RIN of a stabilised 1075 nm fibre amplifier; coating Brownian noise (about $10^{-17}\,\mathrm{m}/\sqrt{\mathrm{Hz}}$); and the mount vibration spectrum at 10 to 40 kHz.